[MP2] testcase func 0 錯誤訊息詢問

4 4

藍楷竣 (LAN, KAI-CHUN)

All Sections

```
[MP2] Slab size: 3
[FILE] fileinit
[SLAB] New kmem_cache (name: file, object size: 504 bytes, at: 0x0000000087f59000, max objects per slab: 8, support in cache obj: 0)
hart 2 starting
hart 1 starting
[FILE] filealloc
[SLAB] Alloc request on cache file
[SLAB] A new slab 0x0000000087f4e000 (file) is allocated
[SLAB] Object 0x0000000087f4e018 in slab 0x0000000087f4e000 (file) is allocated and initialized
init: starting sh
[FILE] filealloc
[SLAB] Alloc request on cache file
[SLAB] Object 0x0000000087f4e210 in slab 0x0000000087f4e000 (file) is allocated and initialized
[FILE] fileclose
[SLAB] Free 0x0000000087f4e210 in slab 0x0000000087f4e000 (file)
[SLAB] End of free
$ mp2
[SLAB] kmem_cache { name: file, object_size: 504, at: 0x0000000087f59000, in_cache_obj: 0 }
[SLAB]  [ partial slabs ]
[SLAB]      [ slab 0x0000000087f4e000 ] { freelist: 0x0000000087f4e210, nxt: 0x0000000000000000 }
[SLAB]          [ idx 0 ] { addr: 0x0000000087f4e018, as_ptr: 0x0000000900000003, as_obj: {scause=0xc sepc=0x0 stval=0x0
panic: kerneltrap
qemu-system-riscv64: terminating on signal 15 from pid 8280 (make)
```

```
== Test public/mp2-0 (3%) ==
public/mp2-0: FAIL (0/3) (3.0s)
    Invalid xv6 state <panic: kerneltrap>, remained debug message to parse: [SlabPrintfKmemStatusData(addr=2281017344, origin='kmem_cache { name: file, object
_size: 504, at: 0x0000000087f59000, in_cache_obj: 0 }', name='file', object_size=504, in_cache_obj=0), SlabPrintfSlabListStatusData(origin='[ partial slabs ]'
, slab_type='partial'), SlabPrintfSlabStatusData(addr=2280972288, origin='[ slab 0x0000000087f4e000 ] { freelist: 0x0000000087f4e210, nxt: 0x0000000000000000
}', freelist=2280972816, nxt=0)]
    QEMU output saved to out/public/mp2-0.out.failed
Score: 0/3
```

從錯誤訊息與 mp2-0.out.failed 可以推測問題應該出在 <as_obj> ，我的作法是 print_fn(obj); 其中 obj的型態是struct run* ，我有嘗試加上(void*)，但輸出還是一樣的，想知道應該要如何修正？

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楊子慶 (YANG, TZU-CHING) (https://cool.ntu.edu.tw/courses/46296/users/229158)

3:25am

同學您好：

同學可能嘗試將不合法的記憶體位置傳入 print_fn 函式，導致核心崩潰。

子慶

(http

藍楷竣 (LAN, KAI-CHUN) (https://cool.ntu.edu.tw/courses/46296/users/184767)

1:00pm

可是我在alloc的時候 0x0000000087f4e018 確實有 object alloc，為甚麼把這個位址傳入 print_fn 函數中就會報錯？

Reply

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林宗聖 (LIN, TZUNG-SHENG) (https://cool.ntu.edu.tw/courses/46296/users/125994)

11:47am

我之前也有遇到類似的問題，

```
== Test public/mp2-0 (3%) ==
public/mp2-0: FAIL (0/3) (2.4s)
    Invalid xv6 state <panic: kerneltrap>, remained debug message to parse: [SlabAllocRequestData(origin='Alloc request on cache file', name='file')]
    QEMU output saved to out/public/mp2-0.out.failed
```

後來我在 kmem_cache_alloc() 裡面初始化 slab 時，

在 slab->freelist = .. 那邊，改成用 page 起始位址，就沒遇到這類問題了，

也許可以供你參考～

(http

藍楷竣 (LAN, KAI-CHUN) (https://cool.ntu.edu.tw/courses/46296/users/184767)

1:12pm

但我的好像是在print的時候出錯的，因為我把 print_fn 函式拿掉之後就可以完整跑完剩下的東西（而且是正確的），所以應該是在 print_fn的時候出錯的，但我在alloc的時候 0x0000000087f4e018 確實有 object alloc，為甚麼把這個位址傳入print_fn 函數中就會報錯？

Edited by 藍楷竣 (LAN, KAI-CHUN) (https://cool.ntu.edu.tw/courses/46296/users/184767) on Apr 3 at 1:13pm

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